

Task 3: **Data Analysis Using Pandas**

Objective

Analyze a dataset, clean the data, and generate summary statistics and insights.

Requirements Completed

- ✓ Loaded CSV file using Pandas
- ✓ Performed data cleaning
- ✓ Applied filtering
- ✓ Performed grouping operations
- ✓ Generated summary statistics
- ✓ Produced meaningful insights

Code:

```
import pandas as pd

# Load CSV file
df = pd.read_csv("sales_data.csv")

print("Dataset Loaded Successfully\n")

# Show first rows
print(df.head())

# Check missing values
print("\nMissing Values:\n", df.isnull().sum())

# Remove duplicates
df.drop_duplicates(inplace=True)

# Convert date column
df["Date"] = pd.to_datetime(df["Date"])

# Create Total Sales column
df["Total_Sales"] = df["Quantity"] * df["Price"]

print("\nCleaned Data:\n", df)
```

```
# Summary statistics
print("\nSummary:\n", df.describe())

# Category wise sales
category_sales = df.groupby("Category")["Total_Sales"].sum()
print("\nCategory Sales:\n", category_sales)

# Best category
print("\nHighest Sales Category:", category_sales.idxmax())

# Total revenue
print("Total Revenue:", df["Total_Sales"].sum())
```

```
import pandas as pd

# Load CSV file
df = pd.read_csv("sales_data.csv")

print("Dataset Loaded Successfully\n")

# Show first rows
print(df.head())

# Check missing values
print("\nMissing Values:\n", df.isnull().sum())

# Remove duplicates
df.drop_duplicates(inplace=True)

# Convert date column
df["Date"] = pd.to_datetime(df["Date"])

# Create Total Sales column
df["Total_Sales"] = df["Quantity"] * df["Price"]

print("\nCleaned Data:\n", df)

# Summary statistics
print("\nSummary:\n", df.describe())

# Category wise sales
category_sales = df.groupby("Category")["Total_Sales"].sum()
print("\nCategory Sales:\n", category_sales)

# Best category
print("\nHighest Sales Category:", category_sales.idxmax())

# Total revenue
print("Total Revenue:", df["Total_Sales"].sum())
```

```

25%    1002.75000    2.75000    2750.000000    2026-01-14 06:00:00    13500.000000
50%    1004.50000    3.50000    11000.000000    2026-01-19 00:00:00    37500.000000
75%    1006.25000    5.25000    26250.000000    2026-01-22 18:00:00    78750.000000
max     1008.00000    10.00000    55000.000000    2026-01-28 00:00:00    110000.000000
std       2.44949     2.66927    18761.796289             NaN    40746.603715

```

```

Category Sales:
Category
Clothing      20000
Electronics   335000
Furniture     29000
Name: Total_Sales, dtype: int64

```

```

Highest Sales Category: Electronics
Total Revenue: 384000

```

Dataset Loaded Successfully

	Order_ID	Customer	Category	Product	Quantity	Price	Date
0	1001	John	Electronics	Laptop	2	55000	2026-01-10
1	1002	Anu	Furniture	Chair	5	3000	2026-01-12
2	1003	Ravi	Electronics	Phone	3	25000	2026-01-15
3	1004	Meena	Clothing	Shirt	10	800	2026-01-18
4	1005	Arun	Furniture	Table	2	7000	2026-01-20

```

Missing Values:
Order_ID      0
Customer      0
Category      0
Product       0
Quantity      0
Price         0
Date          0
dtype: int64

```

```

Cleaned Data:
Order_ID Customer Category ... Price Date Total_Sales
0      1001      John  Electronics ...  55000 2026-01-10    110000
1      1002       Anu   Furniture   ...   3000 2026-01-12     15000
2      1003      Ravi  Electronics ...  25000 2026-01-15     75000
3      1004     Meena   Clothing   ...    800 2026-01-18      8000
4      1005      Arun   Furniture   ...   7000 2026-01-20     14000
5      1006      Kavi  Electronics ...  15000 2026-01-22     60000
6      1007     Latha   Clothing   ...   2000 2026-01-25     12000
7      1008     Rahul  Electronics ...  30000 2026-01-28     90000

```

[8 rows x 8 columns]

```

Summary:
Order_ID Quantity Price Date Total_Sales
count      8.00000    8.00000    8.000000    8      8.000000
mean    1004.50000    4.37500  17225.000000  2026-01-18 18:00:00  48000.000000
min     1001.00000    2.00000    800.000000  2026-01-10 00:00:00    8000.000000

```

Conclusion:

The dataset was successfully cleaned and analyzed using Pandas, and meaningful insights were generated from the data.